

Table A2: Rater 1’s Evaluation of the reviewed studies across five methodological rigour criteria: detection performance, theoretical foundations, computational efficiency, interpretability, and ecological relevance. Each criterion is scored on a scale of 0 (very poor) to 3 (excellent), where scores reflect the specific implementation reported in each study rather than the general capabilities of the underlying method.

Paper	Detection Performance Score	Theoretical Foundation Score	Computational Efficiency Score	Interpretability Score	Ecological Relevance Score
Zaza et al. [30]	2 The paper reports that deep learning autoencoders, particularly NN-AE, outperform traditional models in identifying anomalies, but validation is limited across ecological scenarios.	2 Models are based on established theories but lack deep exploration specific to ecology.	1 Deep learning models are computationally intensive, with limited scalability demonstrated.	2 SHAP and feature-wise reconstruction error analysis provide post-hoc feature importance; mechanistic inference-time explanations absent.	3 Detected anomalies are explicitly linked to ecological events, and the system is positioned for real-time ecosystem monitoring of estuarine fish movement in South African waters, with direct conservation management implications.
Kareinen et al. [31]	3 Extensive validation with metrics like accuracy and F1-score, demonstrating strong detection capability.	2 Based on well-established theories in open-set recognition, but applies existing methods without new theoretical innovation.	1 Deep neural networks are resource-intensive; efficiency not explicitly discussed.	1 Feature-based methods are less transparent; limited interpretability.	3 Addresses ecological monitoring with direct relevance, focusing on plankton recognition.
Siddagangaiah [40]	2 Precision, recall, and F1 scores explicitly reported with confusion matrices across three frequency bands. High false positives in biological ranges due to natural variability being misclassified as anomalies.	3 IF is well-established with strong theoretical basis [56]. Method justified for computational efficiency and accuracy. Validated against known extreme events (typhoons, cold bursts, flooding) with temporal meteorological correlation.	3 Linear time complexity processing 8,568 hourly observations. Optimised to 100 estimators with rapid execution on yearly data. Scalable for real-time marine acoustic monitoring networks.	2 Anomaly scores provide probabilistic interpretation. Detected anomalies visualised with colour coded event types. Integration with Neural-Prophet trend decomposition contextualises anomalies within seasonal patterns, but lacks mechanistic explanations.	3 Detects extreme environmental events (typhoons, storms) and biological disruptions (fish/shrimp vocalisation changes) in passive acoustic data. Clear utility for marine ecosystem management and conservation decision-making in response to climate driven extreme events.
Pu et al. [38]	2 AUROC, AUPR, TNR95 and detection accuracy are reported and used to validate against three baselines, but precision and recall are not reported and failures are not analysed.	2 Based on established deep learning architectures, no new theoretical frameworks.	2 A small Wide ResNet trained for 100 epochs suggests acceptable efficiency, but no runtime, hardware or inference speed is reported.	1 Limited. No saliency or attention analysis is given, and the detector outputs only one score per image.	3 Addresses ecological issues, improves detection of plankton anomalies.
Beyan [33]	2 Recall, true negative rate, geometric mean and AUC are reported, used for validation and analysed in depth, with high accuracy, but the best parameters are taken on the test folds and not on validation data.	2 Builds on established clustering and outlier detection, with some justification for the new filter, classifier and hierarchy.	2 The filter and the classification of a new trajectory are cheap, but training the hierarchy is costly and no runtimes are given.	1 Provides visualisations; decisions are complex.	3 Highly relevant for ecological applications, marine trajectories.

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Table A2: — continued from previous page.

Paper	Detection Performance Score	Theoretical Foundation Score	Computational Efficiency Score	Interpretability Score	Ecological Relevance Score
Walker and Orenstein [34]	2 Outperforms baseline in OOD detection with metrics like 94% accuracy; detailed class-level analysis limited.	2 Builds on established out-of-distribution detection theories; no formal proofs.	2 Uses pre-trained ResNet-18; training manageable, scalable but resource-dependent.	2 Visualisation of class distributions; core models are complex, interpretability moderate.	3 Addresses ecological detection of rare species, important for conservation.
Wang et al. [39]	3 High detection performance with F1 scores up to 98%, validated on real underwater datasets.	2 Based on deep learning principles; no novel theoretical frameworks introduced.	3 Highly efficient, suitable for real-time deployment on resource-constrained devices.	2 Visualisations provided; moderate interpretability, lacks detailed explainability.	3 Addresses ecological monitoring and behaviour recognition, highly relevant.
Chen and Zhu [36]	2 ROC-AUC around 0.98; metrics like precision/recall provided; strong validation overall.	2 Grounded in knowledge distillation and neural network theory; solid but not novel.	2 Lightweight model suitable for deployment; no explicit benchmarks but inference is fast.	2 Gradient-based heatmaps aid interpretability; core decisions remain “black box”.	3 Critical for fisheries management and ecological data reliability.
Beyan et al. [32]	1 Detection and retention rates (52% and 78%) are reported on 1,000 examined trajectories, but 48% of errors are missed and 22% of correct trajectories are removed.	2 Grounded in established probabilistic models like Restricted Boltzmann Machines; no new theoretical derivations.	2 The mcRBM needs no pairwise distance calculations and is designed to scale to 3.6 million trajectories, but training takes 900 to 1,600 epochs and no runtimes are reported.	2 Visualisations of cluster distributions and outliers; approach is somewhat opaque but balances transparency and complexity.	3 Addresses ecological questions on fish activity patterns; highly relevant for ecological and climate change studies.
Ward et al. [37]	1 No ground truth exists for the anomalies, so no precision, recall or F1-score is reported. The authors note a relatively high chance of false negative results.	2 Uses established statistical models; no new theoretical contributions.	1 Computationally intensive; scalability limited.	1 Provides some visualisation; interpretability limited by model complexity.	3 Addresses ecological monitoring, relevant for long-term studies.
Bilik et al. [29]	2 F1-score, precision, recall and AUC are reported and analysed for nine taxa, but the best overall F1 is 0.75 and the decision threshold is set on the test data.	2 Grounded in deep learning principles like autoencoders and CNNs; no new theoretical innovations.	2 Standard deep learning models; practical implementation without detailed runtime analysis.	2 Visualisations of reconstructions and feature clustering; moderate interpretability, internal decisions opaque.	3 Addresses ecological challenges in phytoplankton parasite detection; highly relevant and impactful.
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Paper	Detection Performance Score	Theoretical Foundation Score	Computational Efficiency Score	Interpretability Score	Ecological Relevance Score
Shutler et al. [35]	2 Classifies high biomass algal blooms with 68% accuracy; validation limited by small in situ comparison.	2 Based on established statistical background subtraction techniques; no novel theoretical framework.	3 Pixel-wise statistical calculations; computationally inexpensive and suitable for real-time large scale monitoring.	2 Produces intuitive anomaly maps; transparency in detection criteria but limited explanation of data distribution.	3 Supports early detection of HABs; highly relevant for ecological and health management.
Wu et al. [41]	3 Overall accuracy, average accuracy, F1-score and Kappa are reported on three GF-1 scenes (average overall accuracy 97.66%, average accuracy 84.66%), benchmarked against seven methods and checked for parameter sensitivity and transfer to three further satellites, although reference maps come from visual interpretation and no confidence intervals are given.	2 Grounded in the colour and filament morphology of <i>Noctiluca</i> blooms and standard operators (gamma correction, CLAHE, Otsu thresholding, Isolation Forest), but the four parameters (gamma, window length, spectral ratio and score regulator) are tuned empirically on a single image and no theoretical guarantees are given.	2 Runtime is reported against the comparison methods on a CPU and GPU workstation and described as acceptable, but it varies with how fragmented the bloom is, and no scalability analysis for operational monitoring is provided.	3 Every stage is knowledge driven, with hue maps, line structure energy and spectral rules that map directly onto bloom colour and shape, and the detections added at each verification stage are shown, so outputs are transparent without post hoc explanation tools.	3 It directly targets harmful red tide detection across four satellite platforms, with clear relevance to fisheries, aquaculture and coastal monitoring.
Chuang et al. [42]	2 Precision, a custom interval recall and F1-IR are reported against Monte Carlo baselines with Bonferroni correction, but the best early-detection model reaches only 22.9% precision and 40.7% interval recall, and the data cover six ponds on one farm over ten months with no external validation.	2 Detector choice is justified by stationarity testing and bootstrapped critical values, but residuals are assumed Gaussian without testing and the one-layer, seven-unit LSTM configuration is not justified.	2 A small LSTM on daily data is inexpensive, but no runtime, training time or hardware is reported and real-time use is not addressed.	2 Flagged days are displayed against water colour on each pond's time series, and univariate and multivariate models are compared, but the LSTM offers no feature attribution to show which variables drive a flag.	3 It links temperature and pH anomalies to water colour changes that signal diatom and dinoflagellate growth, giving farmers an early warning with direct practical use, although the setting is farmed ponds rather than open marine ecosystems.
Hammond et al. [43]	1 Only the Calinski-Harabasz clustering index and visual checks are reported, with out-of-sample application but no labelled reference, precision or recall, and extreme event thresholds are not validated against fishing or in situ data.	2 Built on standard K-means, the elbow criterion and peaks over threshold quantiles, consistent with the high chlorophyll and low SST signature of upwelling, but the number of clusters and the quantile thresholds (5/95, 15/85 and 25/75%) are chosen manually rather than by a formal selection procedure.	2 K-means is fitted on a random subset and applied to near real-time daily data, suggesting low cost, but no runtime or scalability figures are reported.	3 Clusters map onto upwelling core, surrounding and non-upwelling waters in chlorophyll and SST space, and extremes are simple historical quantiles, so outputs are transparent and easy to communicate, though cluster boundaries depend on the chosen number of clusters.	3 It directly addresses coastal upwelling and its extremes, which drive productivity and small scale fisheries in the Western Indian Ocean, as the basis for an operational Upwelling Watch service.